

The ML Architecture, validated.



A single reference for the whole engine: what it does, why each ML choice is the right one, how the missing-data problem is actually solved, and where it scales. Written so one person can hold the whole system in their head — and teach it to the next.

00 How to read the diagrams

Every diagram in this document uses one consistent colour language. Colour encodes a component's *role* — and the roles are the whole point of the design. Learn these five and you can read the system at a glance:

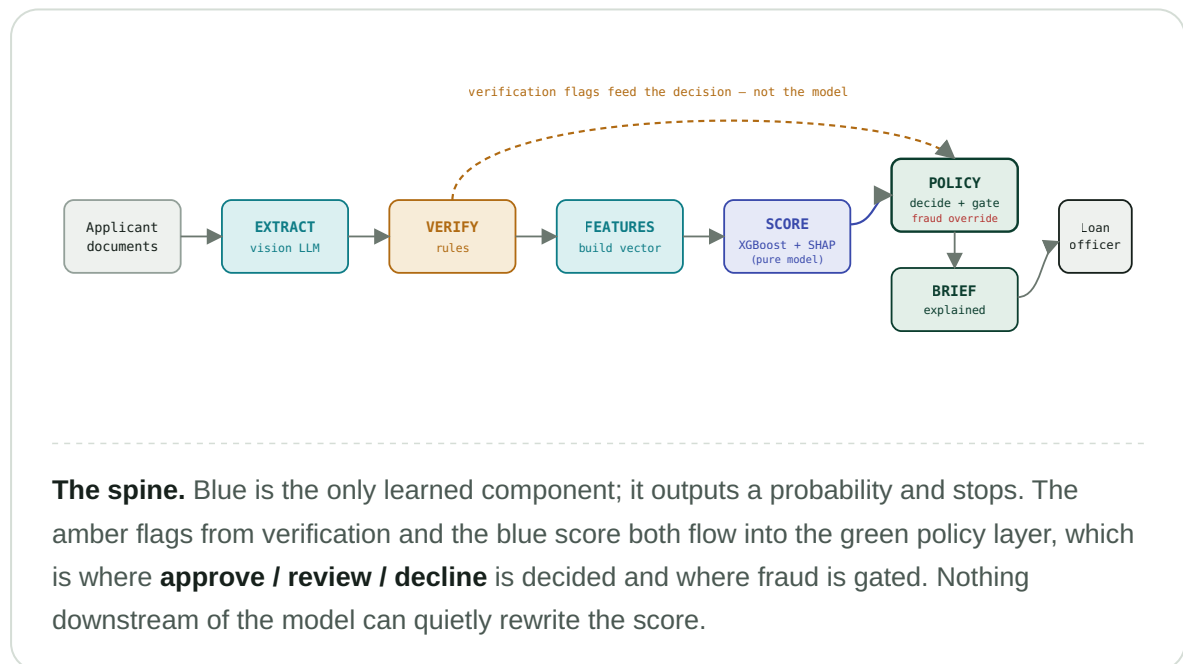
-  AI-read (vision / extraction)
-  Deterministic (rules, policy, thresholds)
-  Learned (the ML model & its explanations)
-  Decision & trust (recommendation, brief, human)
-  Risk / override (fraud, hard stops)

THE GOVERNING PRINCIPLE

The model produces a score. A separate policy layer produces the decision. The model never sees the fraud rules; the rules never change the score. This one separation is what makes the system auditable — hold onto it as you read.

01 The system at a glance

An application flows left to right. Documents are read, checked, turned into features, scored by the model, then — critically — the *pure* score and the verification flags meet in a policy layer that issues the recommendation. A human always makes the final call.



02 Architecture

validation

Stage by stage — is each choice the right one? The verdict is **sound**, with two honest weaknesses that are already being addressed (§08).

SOUND

Extraction — vision LLM. Document reading is a solved, commoditised problem in 2026; a vision model handles varied bank layouts and Urdu without brittle templates. Correct to reuse, not build.

SOUND

Verification — deterministic rules. Fraud and consistency checks are kept as auditable rules, not learned. Exactly right: these must be stable and explainable, and a bank can inspect every one.

SOUND

Scoring — XGBoost + SHAP. Gradient-boosted trees are the empirical state of the art for tabular credit data, and SHAP gives exact per-decision reasons. The pairing is the industry default for a reason (§03).

SOUND

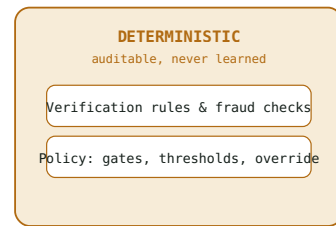
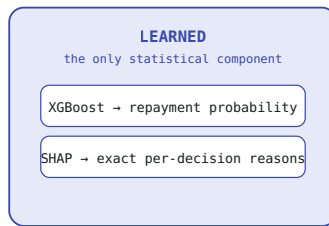
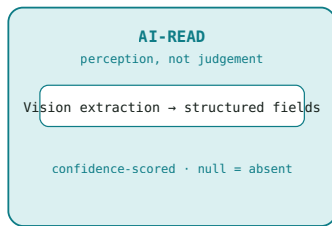
Policy layer — score $\neq$ decision. The strongest single choice in the system. Separating the risk score from the lending decision is how real credit engines are built; it lets appetite and fraud rules change without ever corrupting the model.

CEILING

Training data — synthetic. The pipeline is correct; its ceiling is the quality of the generative assumptions, not real outcomes. Mitigated by anchoring to published figures (§04), but it is the defining limitation.

PARTIAL

Feature richness. Some inputs run on median defaults and features are summary-level, not transaction-level. Real signal is left on the table — the roadmap (§06) closes this.



Three kinds of component, never blurred. Perception (read the document), judgement (score the risk), and policy (decide and gate) are cleanly separated. The model is a single box in the middle — small, pure, and swappable — surrounded by logic a human can audit.

03 Why these ML choices

Why gradient-boosted trees (XGBoost), not a neural network

Credit data is *tabular and heterogeneous* — a mix of amounts, ratios, counts, and categories. On this kind of data, gradient-boosted trees consistently beat deep learning: they capture non-linear interactions, handle mixed and missing values natively, need no feature scaling, and — decisively for us — perform well on *small* datasets. Deep nets want far more data than we have and cost us interpretability. This isn't a fashion choice; the academic benchmarks and the fintech industry both converge here.

Why SHAP, and why it matters legally

Lending decisions must be explainable — a bank has to be able to say *why* it declined someone. SHAP's TreeExplainer gives *exact* Shapley values for tree ensembles (not an approximation), so every decision decomposes into precise per-feature contributions. That is what powers the reason codes and the "here's how to qualify" counterfactuals. Explainability is designed in, not bolted on.

Why the rules stay out of the model

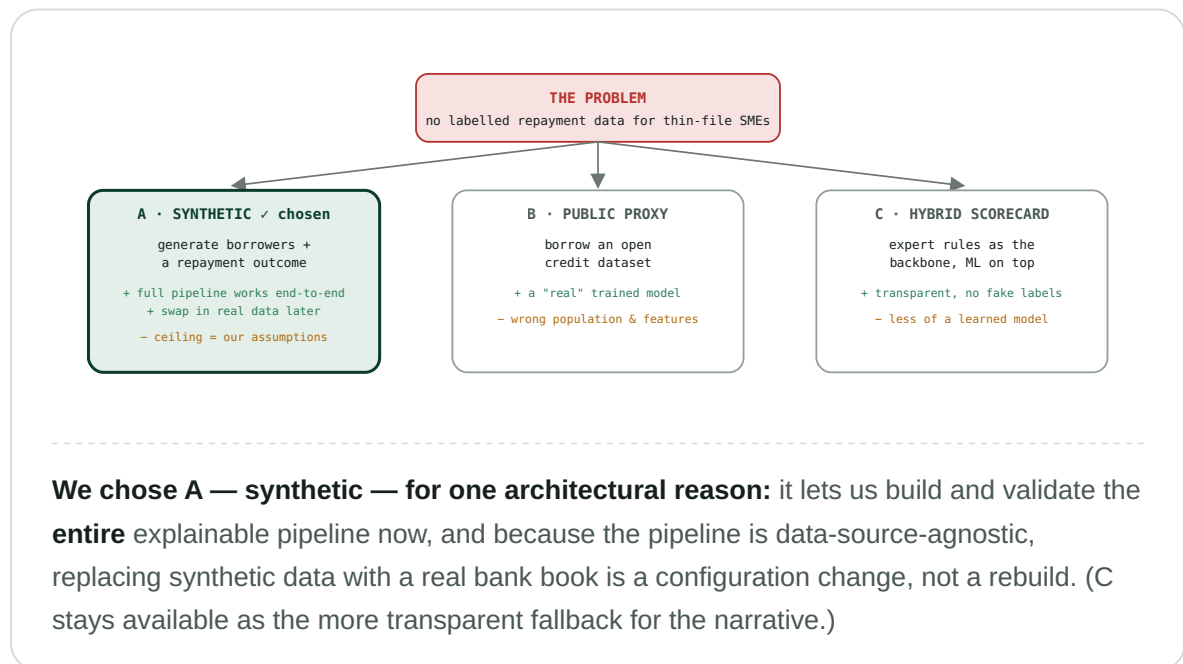
Fraud and consistency checks are moved into a deterministic layer *on purpose*. A name mismatch between a CNIC and a bank statement isn't a probability to be weighed — it's a stop. Keeping it as a rule means it's stable, inspectable, and can't be "learned away" by the model. It also keeps *model risk* and *policy risk* separable, which is standard practice in regulated lending.

THE LOAD-BEARING IDEA

A credit engine is not a score calculator. The model estimates **how likely to repay**; the policy layer decides **what to do about it** given the bank's appetite and the fraud signals. Conflating the two is the most common way these systems become impossible to trust or audit — and we don't.

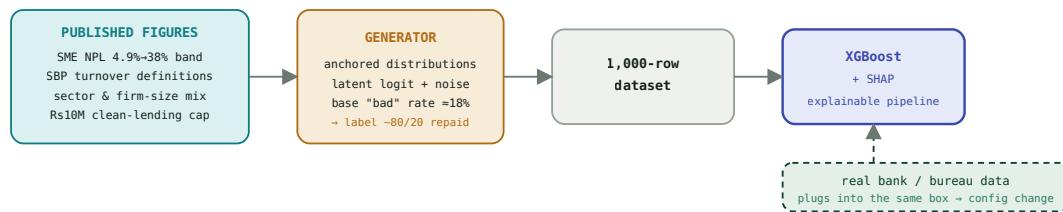
04 The data problem — and how we solve it

Supervised credit scoring needs labelled outcomes: past borrowers, and whether they repaid. For thin-file Pakistani SMEs, **that data does not exist** — which is precisely the market gap Aitabaar addresses. So the honest question isn't "where's your data," it's "what do you do when there is none." There are three routes.



How we make synthetic data honest

Synthetic doesn't mean invented-from-nothing. The generator's distributions and its base default rate are *anchored to published Pakistani figures*, and every parameter is tagged **SOURCED** (a real statistic) or **ASSUMPTION** (a bounded modelling choice) — so "grounded in published data" is literally traceable, line by line.



What it does and doesn't prove. The benchmark shows the pipeline correctly **learns the structure** it was given — it does **not** prove real-world discrimination, because no real outcomes exist to test against. Real data doesn't change the architecture; it sharpens the same model. That honesty is the answer to the judges' hardest question.

The base rate (~18%) is set deliberately between the cleaned portfolio NPL (~4.9%) and the recent stressed peak (~38%), and above the portfolio average because Aitabaar's pool is thin-file and riskier than already-banked firms — a modelling choice bounded by real numbers, not a claimed default rate.

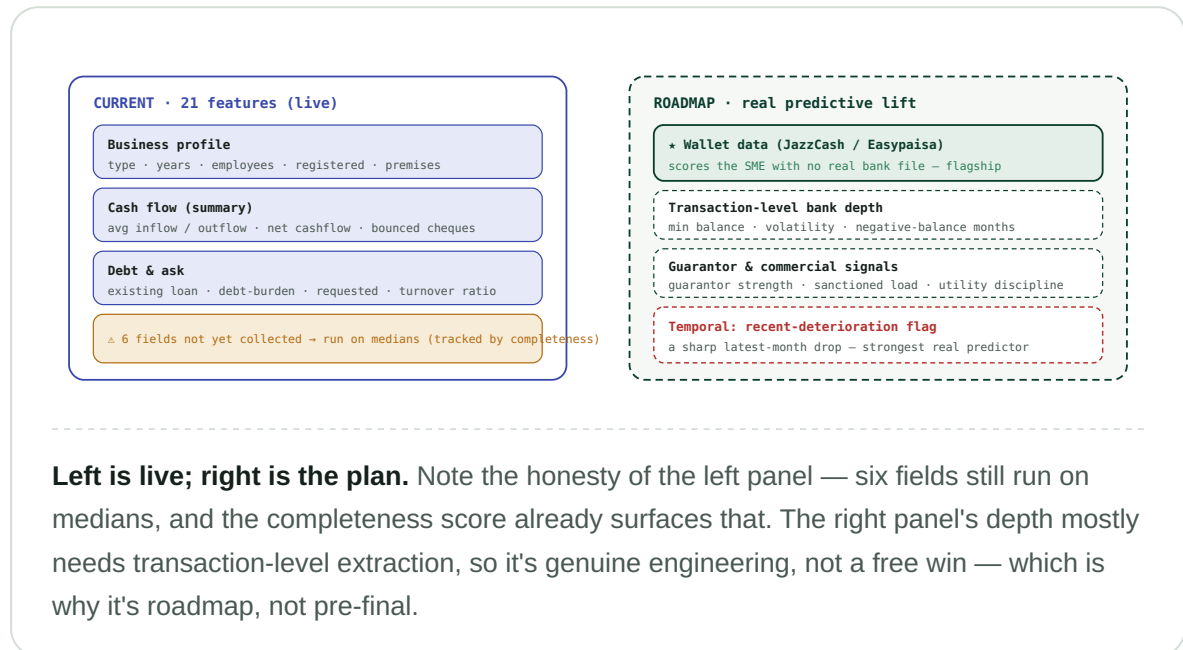
05 What we changed, and why

The engine already worked. The work below hardens it and makes its trustworthiness *visible* — each change earns its place.

- **Decision-policy layer + fraud gate** — turns the score into a recommendation and forces any high-severity flag (e.g. CNIC/bank name mismatch) to REVIEW. Reason: a score-only system would approve fraud with a strong tier; a trust product cannot.
- **Deterministic offline demo** — the demo path runs with no network, no API key, identical every time. Reason: a final can't depend on venue wifi.
- **Data-completeness confidence** — tracks how much of a decision rests on real vs defaulted inputs, and soft-downgrades thin files to REVIEW. Reason: an approve built on guesses shouldn't look like one built on real data.
- **Reason codes + counterfactuals** (next) — adverse-action reasons and a model-verified "reduce bounced cheques to reach Tier B" line. Reason: turns a gatekeeper into an enabler.
- **Anchored synthetic priors** (parallel) — regrounds the generator in SBP/SMEDA figures. Reason: converts "we made up numbers" into "we grounded them."

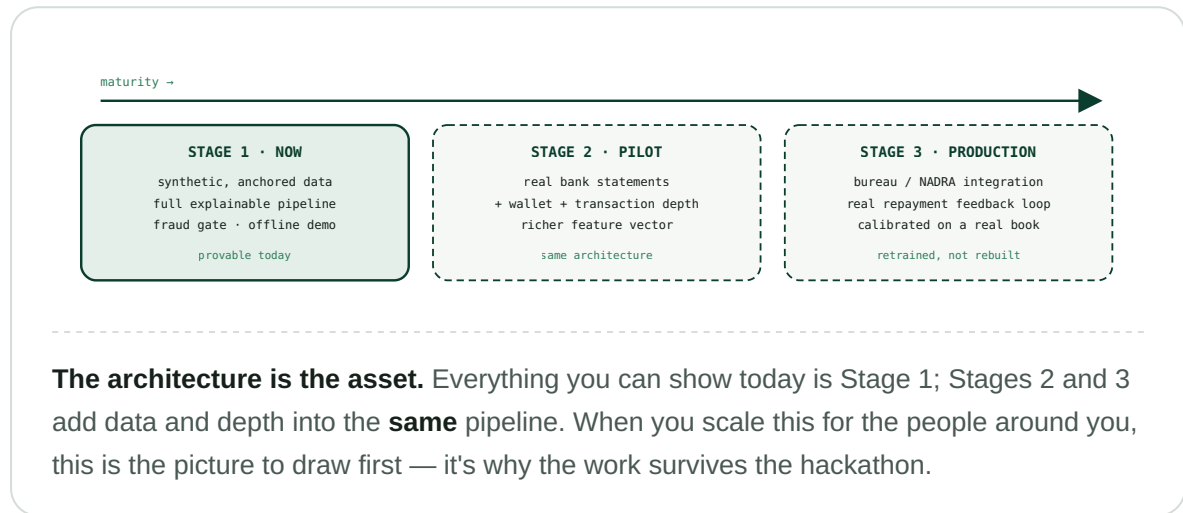
06 Feature architecture — now & next

The model reads a feature vector built from verified data. Today's set is solid but summary-level; the roadmap is where the real predictive lift lives — and the flagship is **wallet data**, which reaches the SME who barely uses a bank at all.



07 How it scales

The reason this is more than a hackathon toy: the architecture is constant across every stage of maturity. Only the *data source* and *feature depth* change — the pipeline, the explainability, and the policy layer stay exactly as they are.



08 Known limitations — state them plainly

Being able to name these *is* the credibility. None is hidden; each has a path.

THE HONEST CAVEATS

- **Synthetic ceiling** — metrics measure recovery of assumed structure, not real discrimination. Closes with real data.
- **Median-defaulted fields** — 6 of 21 inputs aren't collected yet; the completeness score already flags it. Closes with intake expansion.
- **Summary-level features** — the strongest cash-flow signals need transaction-level extraction (roadmap §06).
- **Name-match brittleness** — the fraud threshold can misjudge transliterated names with shared surnames. Harden before relying on judge uploads.
- **No calibration / single split** — probabilities aren't calibrated yet; add calibration + cross-validation before quoting the number as real.
- **In-memory only** — no persistence yet (a deliberate hackathon call).

Bottom line: the architecture is sound and correctly separated, the explainability is real and exact, and the data problem is handled as honestly as it can be without real outcomes. Everything left open is a data-and-depth problem, not a design flaw — which is exactly the position you want to scale from.